

## CEVE 421/521 Schedule

### **i** Note

Check the page for each reading to see discussion questions. You will often be encouraged to focus only on part of relevant papers.

**Final Project:** Climate Plan Audit (*in prep*) — Deliverables are due throughout the semester (see schedule below).

**CEVE 521 students:** You will lead one Paper Discussion session. See Graduate Seminar Requirements.

| Week | Date          | What                                                                                        |
|------|---------------|---------------------------------------------------------------------------------------------|
| 1    | Mon., Jan. 12 | <b>Lecture:</b> Intro to Climate Risk Management slides                                     |
|      | Wed., Jan. 14 | <b>Topical Discussion:</b> T. Frank [1] and S. Crawford [2]                                 |
|      | Fri., Jan. 16 | <b>Lab 1:</b> Julia Setup & Hello World                                                     |
| 2    | Mon., Jan. 19 | <i>Martin Luther King, Jr. Day (No Class)</i>                                               |
|      | Wed., Jan. 21 | <b>Lecture:</b> Hazard, Exposure, Vulnerability slides                                      |
|      | Fri., Jan. 23 | <b>Lab 2:</b> Hazard-Damage Convolution                                                     |
| 3    | Mon., Jan. 26 | <b>Lecture:</b> Climate Variability, Extremes, and Change slides                            |
|      | Wed., Jan. 28 | <b>Paper Discussion:</b> Climate Extremes (IPCC)                                            |
|      | Fri., Jan. 30 | <b>Lab 3:</b> Sea-Level Rise Scenarios and Uncertainty / <b>Project:</b> Topic Proposal Due |
| 4    | Mon., Feb. 2  | <b>Lecture:</b> Uncertainty, Monte Carlo & Risk Metrics slides                              |
|      | Wed., Feb. 4  | <b>Paper Discussion:</b> ICOW Framework slides                                              |
|      | Fri., Feb. 6  | <b>Lab 4:</b> Monte Carlo & Risk Metrics                                                    |
| 5    | Mon., Feb. 9  | <b>Lecture:</b> Valuation & Benefit-Cost Analysis slides                                    |
|      | Wed., Feb. 11 | <b>Paper Discussion:</b> Discounting & BCA                                                  |
|      | Fri., Feb. 13 | <i>No Class</i>                                                                             |

| Week | Date          | What                                                                                                                   |
|------|---------------|------------------------------------------------------------------------------------------------------------------------|
| 6    | Mon., Feb. 16 | <b>Lecture:</b> Optimal Static Decisions slides                                                                        |
|      | Wed., Feb. 18 | <b>Exam 1</b> (Covers Weeks 1-4)                                                                                       |
|      | Fri., Feb. 20 | <b>Lab 5:</b> Simulation-Optimization                                                                                  |
| 7    | Mon., Feb. 23 | <b>Lecture:</b> Value of Information                                                                                   |
|      | Wed., Feb. 25 | <b>Lecture:</b> Value of Imperfect Information / <b>Lecture:</b> Global Sensitivity Analysis slides                    |
|      | Fri., Feb. 27 | <b>Lab 6:</b> Value of Partial Information ( <i>in prep</i> )                                                          |
| 8    | Mon., Mar. 2  | <b>Lecture:</b> Robustness & Deep Uncertainty slides / <b>Project:</b> Teams & Topics Assigned                         |
|      | Wed., Mar. 4  | <b>Paper Discussion:</b> Robustness & Deep Uncertainty ( <i>in prep</i> )                                              |
|      | Fri., Mar. 6  | <b>Lab 7:</b> Robustness Metrics ( <i>in prep</i> )                                                                    |
| 9    | Mon., Mar. 9  | <b>Workshop:</b> Final Project Work Session                                                                            |
|      | Wed., Mar. 11 | <b>Lecture:</b> Trade-offs & Values slides ( <i>in prep</i> ) / <b>Reading:</b> Trade-offs & Values ( <i>in prep</i> ) |
|      | Fri., Mar. 13 | <b>Exam 2</b> (Covers Weeks 5-8)                                                                                       |
|      | Mar. 16-20    | <i>Spring Break</i>                                                                                                    |
| 10   | Mon., Mar. 23 | <b>Lecture:</b> Exploratory Modeling & Scenario Discovery slides ( <i>in prep</i> )                                    |
|      | Wed., Mar. 25 | <b>Paper Discussion:</b> Exploratory Modeling & Scenario Discovery ( <i>in prep</i> )                                  |
|      | Fri., Mar. 27 | <b>Lab 8:</b> Scenario Discovery ( <i>in prep</i> ) / <b>Project:</b> Memo 1 Due                                       |
| 11   | Mon., Mar. 30 | <b>Lecture:</b> Sequential Decision Making slides ( <i>in prep</i> )                                                   |
|      | Wed., Apr. 1  | <b>Paper Discussion:</b> Sequential Decision Making ( <i>in prep</i> )                                                 |
|      | Fri., Apr. 3  | <b>Lab 9:</b> Real Options & Dynamic Policy Search ( <i>in prep</i> )                                                  |
| 12   | Mon., Apr. 6  | <b>Lecture:</b> Multi-Objective Optimization slides ( <i>in prep</i> )                                                 |
|      | Wed., Apr. 8  | <b>Paper Discussion:</b> Multi-Objective Optimization ( <i>in prep</i> )                                               |

| Week   | Date          | What                                                                                      |
|--------|---------------|-------------------------------------------------------------------------------------------|
| 13     | Fri., Apr. 10 | <b>Lab 10:</b> Pareto Fronts ( <i>in prep</i> ) / <b>Project:</b> Memo 2 Due              |
|        | Mon., Apr. 13 | <b>Lecture:</b> Case Studies slides ( <i>in prep</i> )                                    |
|        | Wed., Apr. 15 | <b>Paper Discussion:</b> Case Studies ( <i>in prep</i> )                                  |
| 14     | Fri., Apr. 17 | <b>Project:</b> Slides Due ( <i>in prep</i> )                                             |
|        | Mon., Apr. 20 | Final Project Presentations ( <i>in prep</i> ) (Teams 1 & 2)                              |
|        | Wed., Apr. 22 | Final Project Presentations ( <i>in prep</i> ) (Teams 3 & 4)                              |
| Finals | Fri., Apr. 24 | <i>Backup/overflow day</i>                                                                |
|        | TBD           | <b>Exam 3</b> (Covers Weeks 9-13) / <b>Project:</b> Written Report Due ( <i>in prep</i> ) |

## Bibliography

- [1] T. Frank, "Bold New Jersey Shore Flood Rules Could Be Blueprint for Entire U.S. Coast," *Scientific American*, Aug. 2022, Accessed: Jan. 11, 2023. [Online]. Available: <https://www.scientificamerican.com/article/bold-new-jersey-shore-flood-rules-could-be-blueprint-for-entire-u-s-coast/>
- [2] S. Crawford, "Floods and Storms Are Ravaging the Jersey Shore. Why Do We Keep Building It Back?." Accessed: Dec. 05, 2025. [Online]. Available: <https://www.rollingstone.com/culture/culture-features/floods-storms-jersey-shore-beach-rebuilding-1235477927/>